

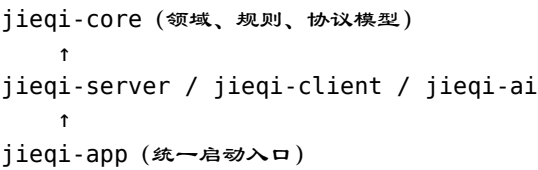
align(center + horizon)[block(width: 15cm, inset: (y: 1.1cm), stroke: (top: 2.5pt + rgb(26, 54, 93),
bottom: 0.75pt + rgb(203, 213, 225)),) [align(center)[text(size: 10.5pt, tracking: 0.35em, fill:
rgb(71, 85, 105))[大 作 业] v(0.55cm) text(size: 26pt, weight: “bold”, fill: rgb(15, 23, 42))[揭棋对弈
程序设计] v(0.65cm) text(size: 19pt, weight: “medium”, fill: rgb(30, 64, 175))[总体架构] v(0.35cm)
if subtitle != “” [text(size: 13pt, fill: rgb(100, 116, 139))[System Architecture — 模块依赖、数据
流、类图] v(0.35cm)]]]

v(1.5cm)

box(width: 13cm, inset: (x: 1.2cm, y: 0.95cm), fill: rgb(248, 250, 252), radius: 6pt, stroke: 0.75pt +
rgb(226, 232, 240),) [align(center)[text(size: 11pt, weight: “bold”, fill: rgb(51, 65, 85))[小组成员]
v(0.55cm) grid(columns: (1fr, 1fr), column-gutter: 1.6cm, row-gutter: 0.65cm, align: center +
horizon, [text(size: 13pt, weight: “bold”)[张恒基 (组长)]
text(size: 12pt, fill: rgb(100, 116, 139))[2024211301 / 2024210926]], [text(size: 13pt, weight: “bold”)
[秦博宇]
text(size: 12pt, fill: rgb(100, 116, 139))[2024211302 / 2024210940]], [text(size: 13pt, weight: “bold”)
[陈艺博]
text(size: 12pt, fill: rgb(100, 116, 139))[2024211302 / 2024210931]], [text(size: 13pt, weight: “bold”)
[陈雨飞]
text(size: 12pt, fill: rgb(100, 116, 139))[2024211301 / 2024210918]],)]]

v(1cm) text(size: 12pt, fill: rgb(100, 116, 139))[技术设计 · 2026-06-18] v(0.5cm) text(size: 10.5pt,
fill: rgb(148, 163, 184))[北京邮电大学 · 计算机学院]]

模块依赖



模块职责

模块	职责	不做什么
jieqi-core	棋盘、棋子、规则、终局、棋谱、协议	不含网络、AI、UI
jieqi-server	WS/TCP 服务、房间、匹配、持久化	不含规则判断
jieqi-client	控制台交互、棋盘显示、命令解析	不含规则判断
jieqi-ai	搜索、评估、Agent 编排、三档 Bot	不含领域逻辑
jieqi-app	统一启动菜单、CLI 参数	无新业务逻辑

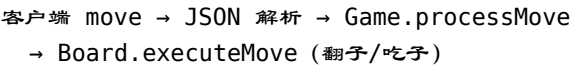
对局主流程



核心类图

Board	10×9 棋盘，棋子查询，make/undo 走子
ChessPiece	color, type, virtualType, revealed, row, col
Move	source, destination, isFlipOnly
Game	对局状态机：board, currentTurn, status, record, replayTimeline
RuleValidator	isValidMove / isMoveLegal / generateStrictLegalMoves
EndgameJudge	checkAfterMove：将死/困毙/和棋/长将/长捉
GameRecord	文字棋谱 → *.jieqi
ReplayTimeline	List<ReplayFrame> 快照时间线
AiBot	接口 ← EasyRuleBot / AlphaBetaBot / BeliefAlphaBetaBot
OptimizedAlphaBeta	搜索内核：AB 剪枝 + 迭代加深 + TT + 静态搜索

数据流



- GameRecord 追加棋谱行
- ReplayTimeline 追加快照帧
- EndgameJudge 终局判定
- JsonMessages 构建 moveResult/gameOver
- WsGameServer 广播给双方

Maven 多模块说明

- 父 POM: pom.xml 聚合 5 模块
- 依赖方向单向: core ← server/client/ai ← app
- 编译策略: mvn package -pl jieqi-app -am 自动构建依赖链
- Fat JAR: jieqi-app/target/unveil-jieqi.jar